

Ankush Yadav

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WORK EXPERIENCE

Associate Software Engineer (Intern)

Feb 2026 – Apr 2026

Mphasis Ltd

Bengaluru, Kar

- Developed responsive web application modules using HTML5, CSS3, Bootstrap, and TypeScript, improving frontend usability and performance.
- Built and integrated backend functionalities using Java, Spring, and Spring Boot for scalable application development.
- Managed database operations with Hibernate and MySQL while using Git, GitHub, and Maven for efficient development workflows.

Data Science Intern

Jul 2025 – Aug 2025

Evoastra Ventures Pvt Ltd

Remote

- Extracted structured data from dynamic websites using BeautifulSoup & Selenium, enabling automated analytics workflows.
- Developed a deep learning image captioning model (CNN+LSTM with DenseNet201 & custom tokenizer) deployed via Flask + HTML/CSS, achieving 85% BLEU score accuracy.
- Improved caption generation performance by 15% through advanced feature extraction and optimized model training.

PROJECTS

AI Multi-Agent Social Debate System | *Python, LangChain, LangGraph, FAISS, Sentence-Transformers, Groq API*

- Developed an AI-powered multi-agent debate platform using LangGraph and FAISS for semantic agent routing and dynamic response generation.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline with conversation memory, semantic retrieval, and prompt injection protection for secure AI interactions.
- Designed a modular multi-agent architecture enabling context-aware, scalable, and intelligent debate simulations across multiple AI personas.

Containerized Flask REST API with Kubernetes & MongoDB | *Python, Flask, Docker, Kubernetes, MongoDB*

- Built and containerized a Flask REST API using Docker and deployed it on a Kubernetes (Minikube) cluster with a 2-replica deployment.
- Configured MongoDB using StatefulSet, Persistent Volume Claims (PVC), Kubernetes Secrets, and internal DNS-based service discovery.
- Implemented Horizontal Pod Autoscaling (2–5 replicas) with CPU resource requests and limits, validating deployment through API testing and Kubernetes monitoring.

Image Captioning Web Application | *Python, TensorFlow, Flask, HTML, CSS*

- Developed a deep learning-based image captioning system using a CNN-LSTM architecture with DenseNet201 for feature extraction.
- Built an end-to-end Flask web application enabling users to upload images and generate descriptive captions in real time.
- Improved caption generation performance through optimized feature extraction and model training, achieving an 85% BLEU score.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, JavaScript, SQL

Frontend Technologies: HTML5, CSS3, Bootstrap, React.js

Backend Technologies: Flask, Spring Boot, REST APIs

DevOps & Cloud: Docker, Kubernetes, Minikube, Git, GitHub, Google Cloud Platform

AI/ML Tools: TensorFlow, Keras, Scikit-Learn, LangChain, LangGraph, FAISS, NumPy, Pandas

Database Technologies: MySQL, MongoDB, PyMongo

Core CS Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Soft Skills: Problem-Solving, Communication, Leadership, Team Collaboration

EDUCATION

Presidency University

Bachelor of Technology in Computer Science

Bengaluru, Kar

Dec 2022 – May 2026

Kendriya Vidyalaya AFS Yelahanka

CBSE Class XII, Stream: Computer Science

Bengaluru, Kar

Apr 2021 – Jul 2022

CERTIFICATIONS

AWS Certified Cloud Practitioner: Amazon Web Services (AWS)

GenAI Studio: Powered by Google Cloud

Networking Basics: Cisco

ACHIEVEMENTS

- Selected in the **top 70 teams out of 450** in the World's Largest Innovation Expo (featuring over 2,600 students) held at Presidency University — *May 2023*
- Built an **Arduino-based four-legged spy robot** equipped with obstacle detection and wireless monitoring capabilities.